Q2. (a) $(1001101010)_2 = 2^0\times0 + 2^1\times1 + 2^2\times0 + 2^3\times1 + 2^4\times0 + 2^5\times1 + 2^6\times1 + 2^7\times0 + 2^8\times0 + 2^9\times1$

$= 2^1 + 2^3 + 2^5 + 2^6 + 2^9 = \boxed{618}$

(b) $(490)_{10} \rightarrow$ octal

8	490	
8	61	2
8	7	5
	0	7

↑ 752

$\boxed{(490)_{10} = (752)_8}$

(c) $(576)_8$ to Hexadecimal

$(576)_8 = 6\times8^0 + 7\times8^1 + 5\times8^2 = 6+56+320 = (382)_{10}$

$(382)_{10}$ to hexa.

16	382	
16	23	14 = E
16	1	7
	0	1

↑

$\boxed{(382)_{10} = (17E)_{16}}$

$\boxed{\text{Ans: } (17E)_{16}}$

(d) ~~$(B9C0)_{16}$~~

~~B: 11 C: 12~~

~~$\Rightarrow$ 11 9 12 0~~

~~$(B9C0)_{16} = (11\times16^3 + 9\times16^2 + 12\times16^1 + 0\times16^0)_{10} = (47552)_{10}$~~

(d) B → 1011

9 → 1001

C → 1100

0 → 0000

$\Rightarrow \boxed{(B9C0)_{16} = (1011\,1001\,1100\,0000)_2}$

(e) $(6537)_8$

$\boxed{(6537)_8 \rightarrow (110101011111)_2}$

6 → 110

5 → 101

3 → 011

7 → 111

(f) $(445)_{10}$

8	445	
8	55	5
8	6	7
	0	6

↑

$(445)_{10} = (675)_8$

(g) $(11001)_2 = 2^0 \times 1 + 2^1 \times 0 + 2^2 \times 0 + 2^3 \times 1 + 2^4 \times 1$

$= 1 + 8 + 16 = 25$

(h) $(4AD)_{16} = 13 \times 16^0 + 10 \times 16^1 + 4 \times 16^2$

A → 10

D → 13

$= 13 + 160 + 1024 = 1197$

$(4AD)_{16} = (1197)_{10}$